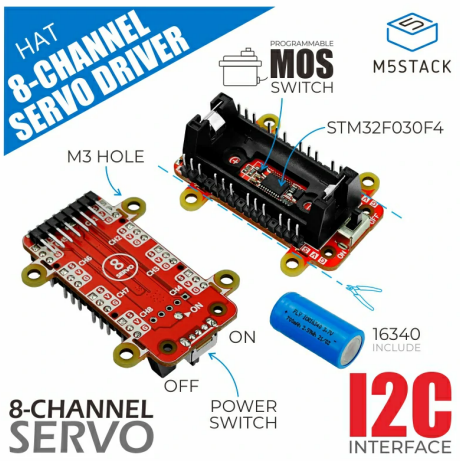
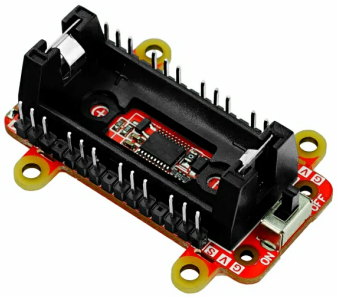
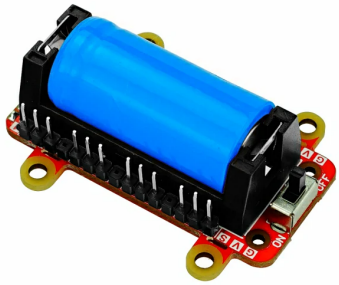
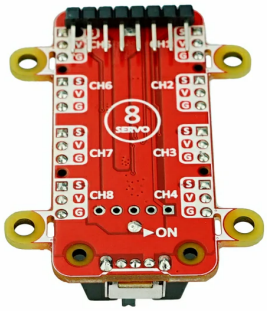
Hat 8Servos v1.1

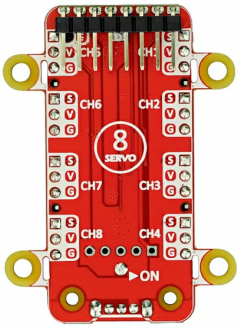
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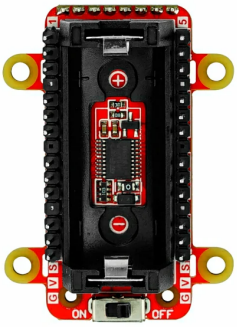












Description

Hat 8Servos v1.1 is an 8-channel servo driver board designed for the M5Stick series. It uses an STM32F030F4 main control to generate multiple PWM signals for servo driving, controlled via the I2C communication protocol to effectively save IO resources. It integrates a MOSFET-based servo power control circuit, supporting programming for dynamic motor power release/lock. Power is provided by rechargeable 16340/18350 lithium batteries (700mAh), also compatible with 18350 lithium batteries. With 8 channels running simultaneously, the maximum load can reach **1.3A**, meeting the driving requirements of standard servo specifications.

Features

- 8-channel servo drive
- Programmable motor power control
- Powered by 16340/18350 lithium battery
- I2C protocol control (0x36)
- Battery reverse protection

Includes

- 1 x Hat 8Servos v1.1
- 1 x 16340/18350 Battery (700mAh)

Applications

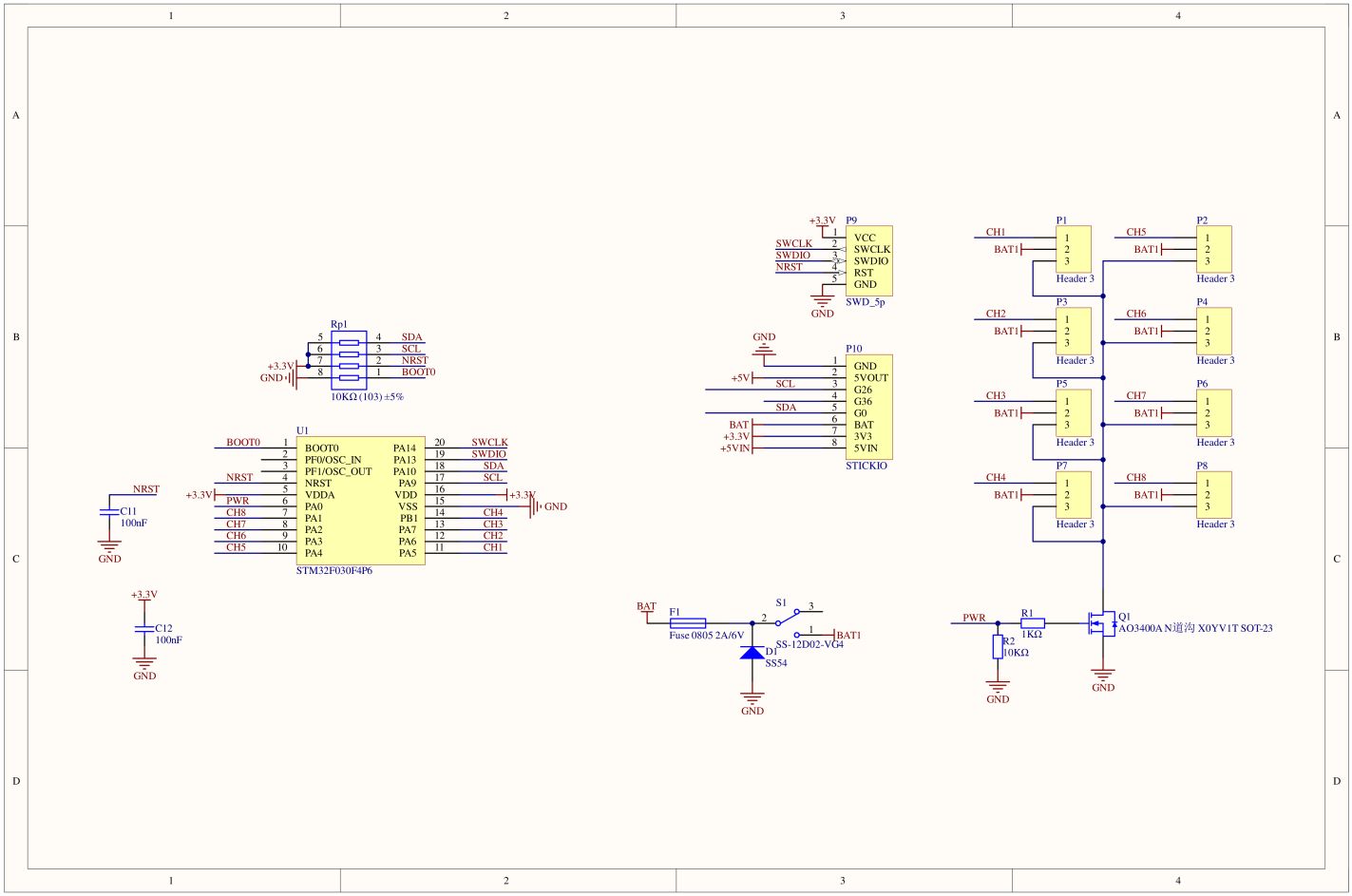
- Servo controller
- Robot control
- Smart toys

Specifications

Specification	Parameter
MCU	STM32F030F4P6
Lithium Battery	Specification: 16340 / 18350, Capacity: 700mAh
Servo Driver Channels	8 Channels
Max Load Capacity per Channel	Max Load Capacity for 8 Channels: DC 4.2V@1.3A
No-load Standby Current	DC 4.2V@2.2uA
Mounting Hole Specification	M3
Communication Interface	I2C Communication @0x36
Product Size	52.0 x 24.0 x 25.0mm
Product Weight	28.3g
Package Size	75.0 x 46.0 x 29.0mm
Gross Weight	39.7g

Schematics

- Hat 8Servos v1.1 Schematics PDF

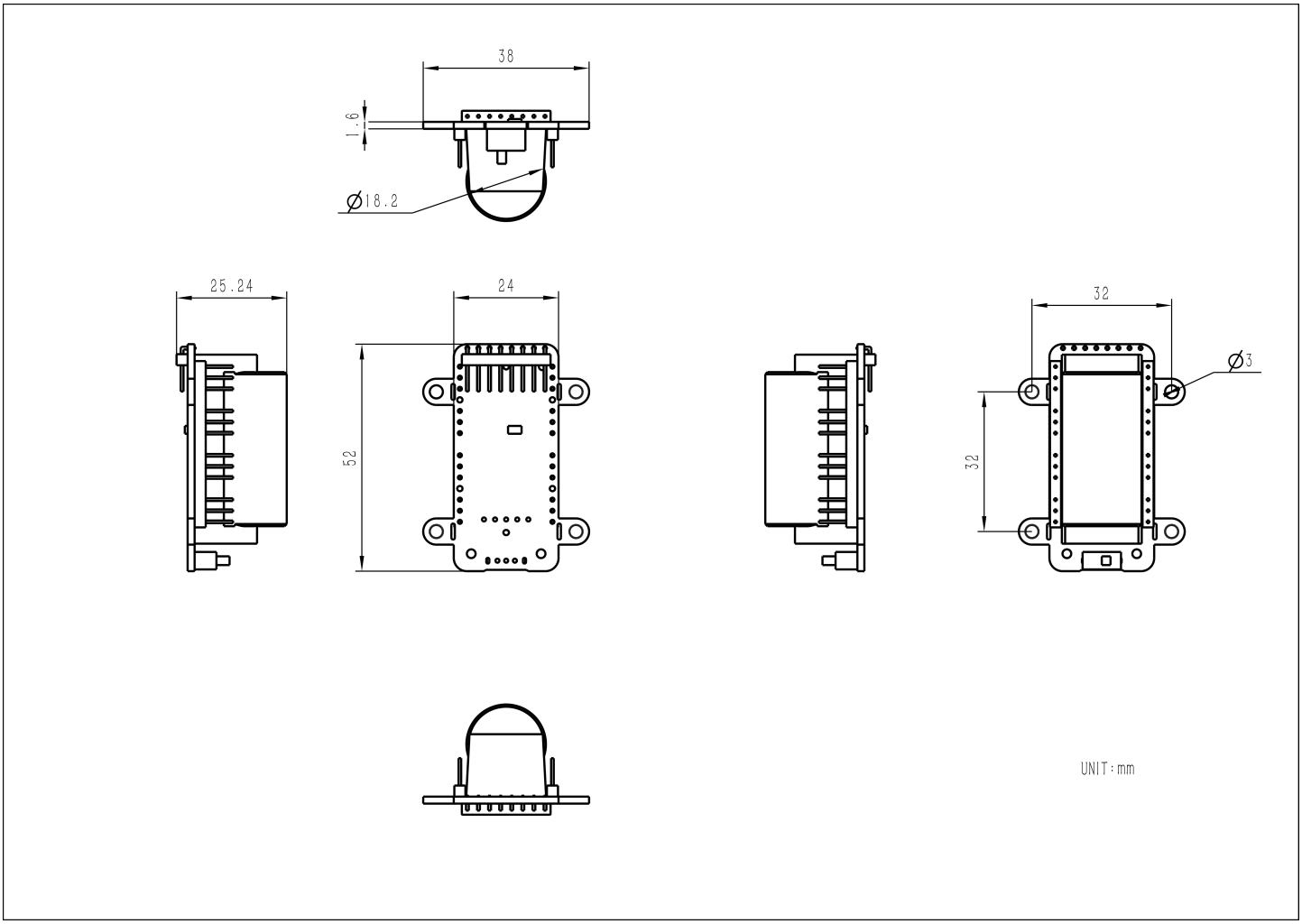


PinMap

M5StickC	G0	G26	3.3V	GND
8Servos HAT v1.1	SDA	SCL	VIN	GND

Model Size

- Hat 8Servos v1.1 Model Size PDF



Softwares

Arduino

- Hat 8Servos v1.1 Arduino Library
- Hat 8Servos v1.1 Example - with M5StickC
- Hat 8Servos v1.1 Example - with M5StickC-Plus

UiFlow1

- Hat 8Servos v1.1 UiFlow1 Docs

Internal Firmware

- Hat 8Servos v1.1 Internal Firmware

Protocol

- Protocol type: I2C
- I2C Address: **0x36**

hex	len	R/W	description	send params
0x00	1	R/W	CH1 angle output	[0] CH1 angle Valid range: 0-180
0x01	1	R/W	CH2 angle output	[0] CH2 angle Valid range: 0-180
0x02	1	R/W	CH3 angle output	[0] CH3 angle Valid range: 0-180
0x03	1	R/W	CH4 angle output	[0] CH4 angle Valid range: 0-180
0x04	1	R/W	CH5 angle output	[0] CH5 angle Valid range: 0-180
0x05	1	R/W	CH6 angle output	[0] CH6 angle Valid range: 0-180
0x06	1	R/W	CH7 angle output	[0] CH7 angle Valid range: 0-180
0x07	1	R/W	CH8 angle output	[0] CH8 angle Valid range: 0-180
0x10	2	R/W	CH1 pulse width output	[0] CH1 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x12	2	R/W	CH2 pulse width output	[0] CH2 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x14	2	R/W	CH3 pulse width output	[0] CH3 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x16	2	R/W	CH4 pulse width output	[0] CH4 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x18	2	R/W	CH5 pulse width output	[0] CH5 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x1A	2	R/W	CH6 pulse width output	[0] CH6 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x1C	2	R/W	CH7 pulse width output	[0] CH7 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x1E	2	R/W	CH8 pulse width output	[0] CH8 pulse width HB [1] CH1 pulse width LB Valid range: 500-2500
0x30	1	R/W	MOS servo power control	[0] MOS_CTL Valid range: 0 (power off)/1 (power on)

Product Comparison

Specification	8Servos HAT	8Servos HAT v1.1
Motor Power Control	/	MOSFET control power on/off
Battery Reverse Protection	/	Protection circuit
Mounting Ears	/	4x removable ears with mounting holes
Programmable RGB LED	SK6812	/
I2C ADDR	0x38	0x36